

Depth over Fidelity in Fixed-Budget Noisy Evolution Strategies

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ICML
International Conference
On Machine Learning



Paper

Code

5–10×

more updates than
k-fold resampling

77–86%

win-rate vs evaluation-
stage denoising

≤ 1

extra eval / candidate
(capped overhead)

30 + 6

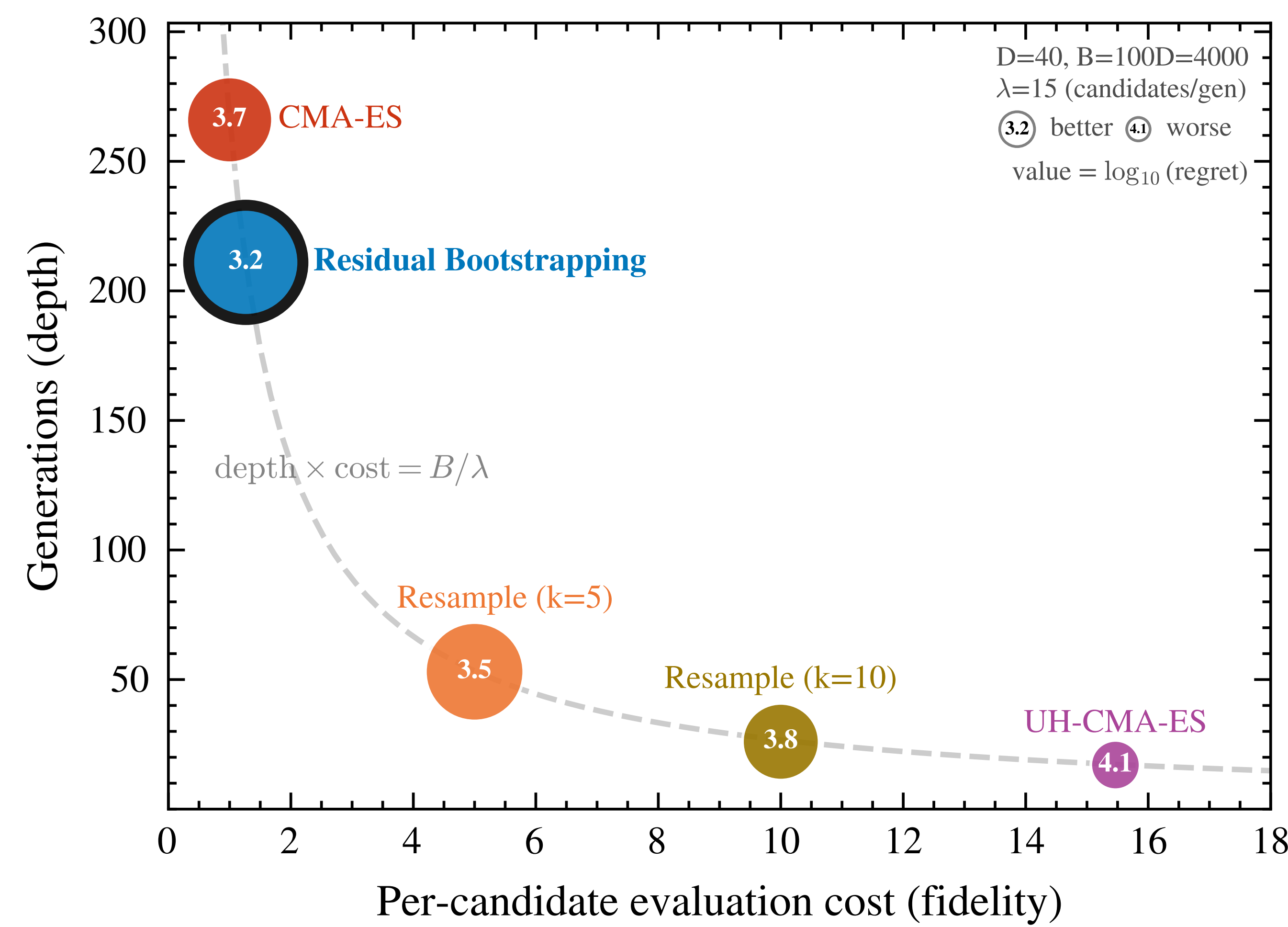
COCO functions ×3 dims
+ external tasks

Background & Motivation

- Noisy oracle $y = f(x) + \varepsilon$; CMA-ES ranks are invariant, but noise permutes them
- “Misranked elites” bias the update $\Rightarrow$ selection noise
- Mainstream fix: spend evaluations to denoise ranks (resampling, racing, UH-CMA-ES)
- But under fixed B , denoising $k \times$ collapses depth to $T \approx B/(k\lambda)$ — fewer mean & covariance steps

How much progress per resample is actually gained by buying fidelity?

The Depth–Fidelity Trade-off



Fixed B : fidelity (per-candidate cost) vs depth (generations). RB-PEM owns the low-cost / high-depth corner.

Contributions

1. Formalize the depth–fidelity trade-off
2. PEM: Rao–Blackwellization of the noisy rank update
3. RB-PEM + Probe-and-Switch, capped overhead
4. Consistent gains across COCO + 6 external tasks

Probabilistic Elite Membership

$$w_{t,i}^* = \mathbb{E}[w(r_{t,i}) \mid x_{t,1:\lambda}]$$

$$w_{t,i}^* = \frac{1}{\mu} \Pr(r_{t,i} \leq \mu \mid x_{t,1:\lambda})$$

- Conditional expected rank weight replaces hard ranks
- Preserves the mean update, cuts its dispersion

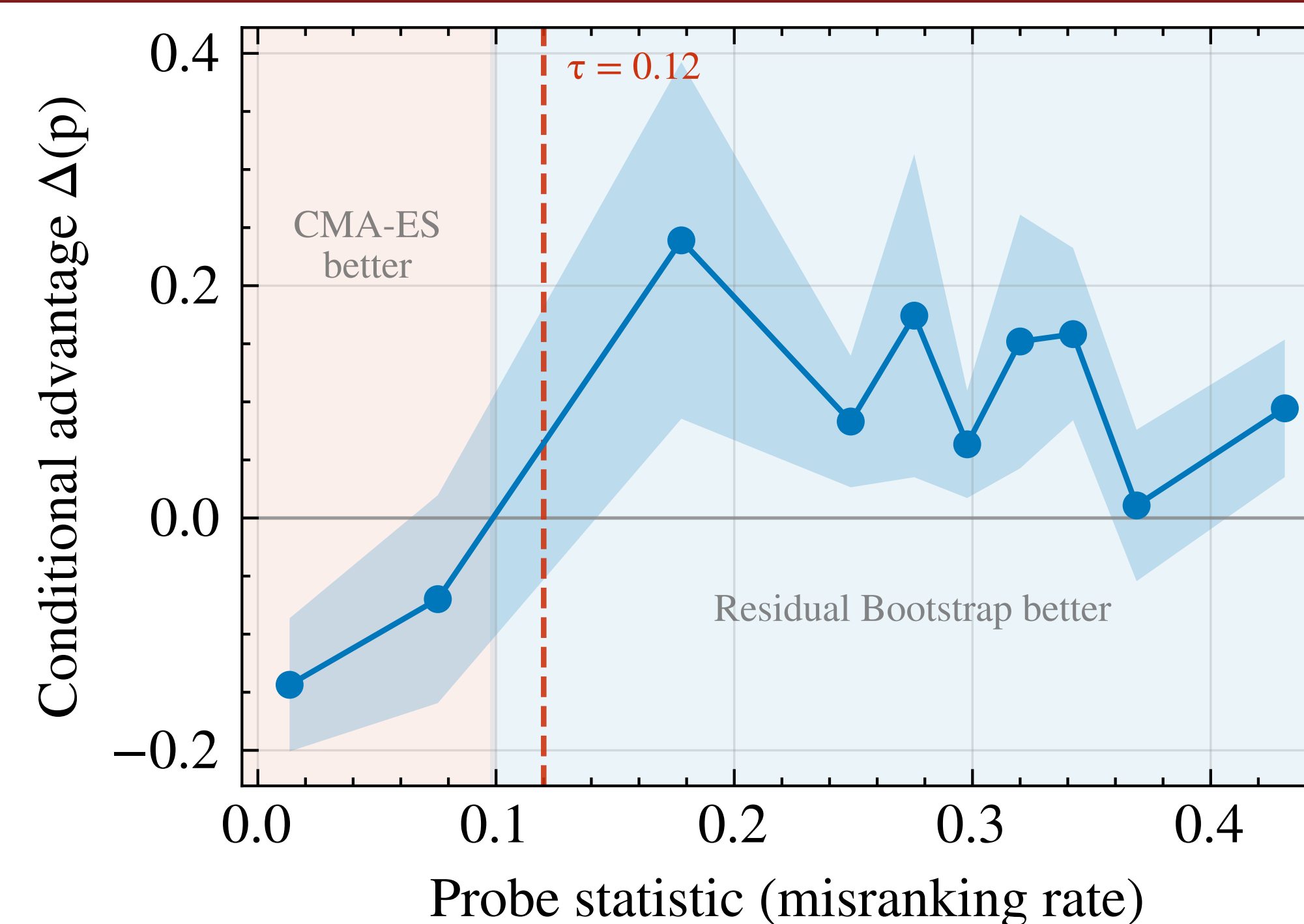
RB-PEM + Probe-and-Switch

$$C_t = \lambda + K_t, \quad K_t \leq K_{\max} \ll \lambda \Rightarrow T \approx B/\lambda$$

- Residual bootstrap simulates many rankings at $\sim$ zero cost

- Probe P : run RB-PEM if $P \geq \tau$ ($=0.12$), else CMA-ES

Single Crossing $\rightarrow$ Switch



Conditional advantage $\Delta(p)$ crosses zero once at $\tau=0.12$ — a Bayes-optimal switch rule.

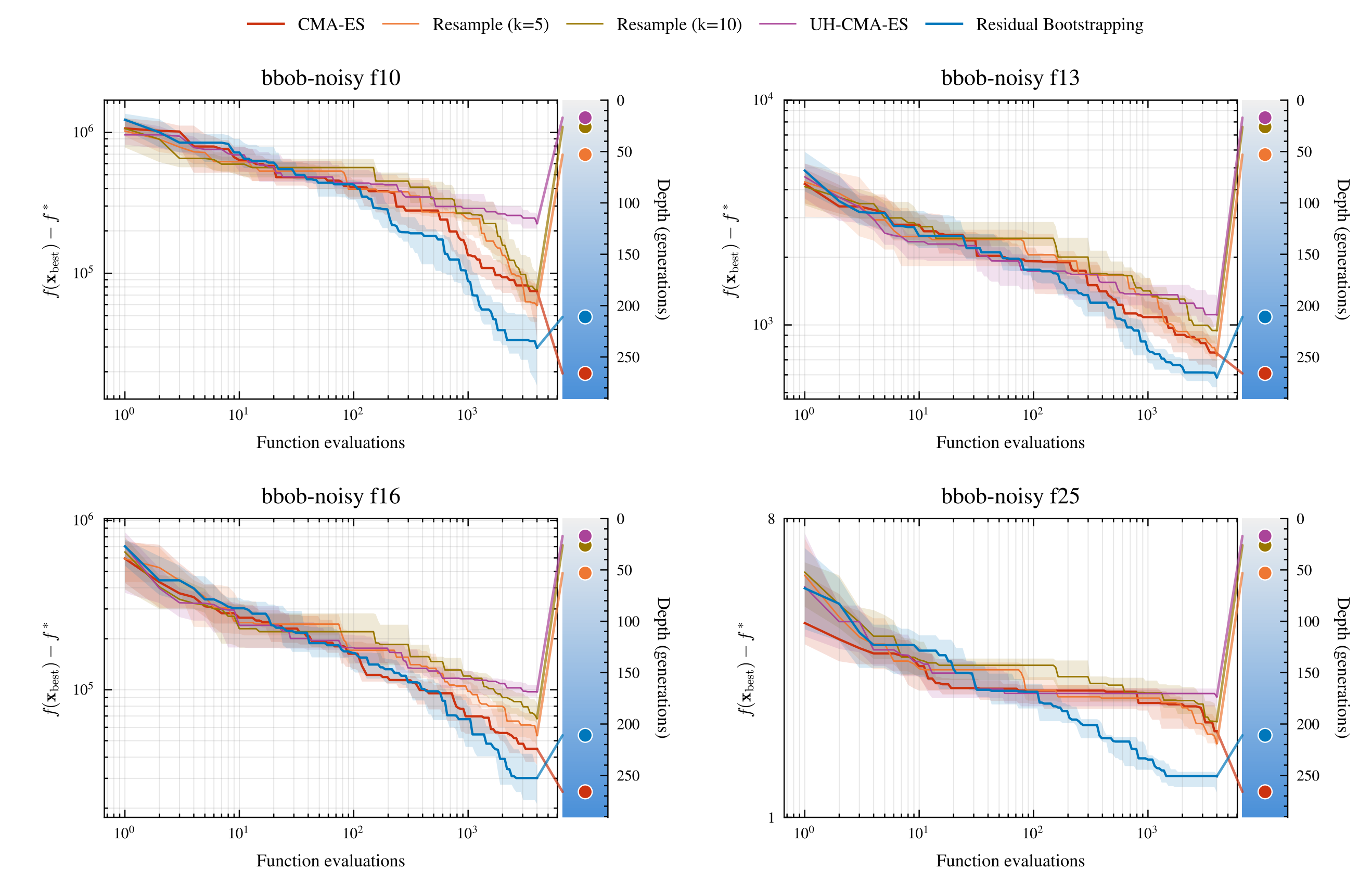
Theory: Why Depth Wins

- Lemma 1 (Rao–Blackwell): PEM is the conditional mean of the noisy update — same mean, lower dispersion

$$\mathbb{E}[f(X) \mid x_{1:\lambda}] \geq f(\bar{X}) + \frac{\sigma}{2} \mathbb{E}\|X - \bar{X}\|^2$$

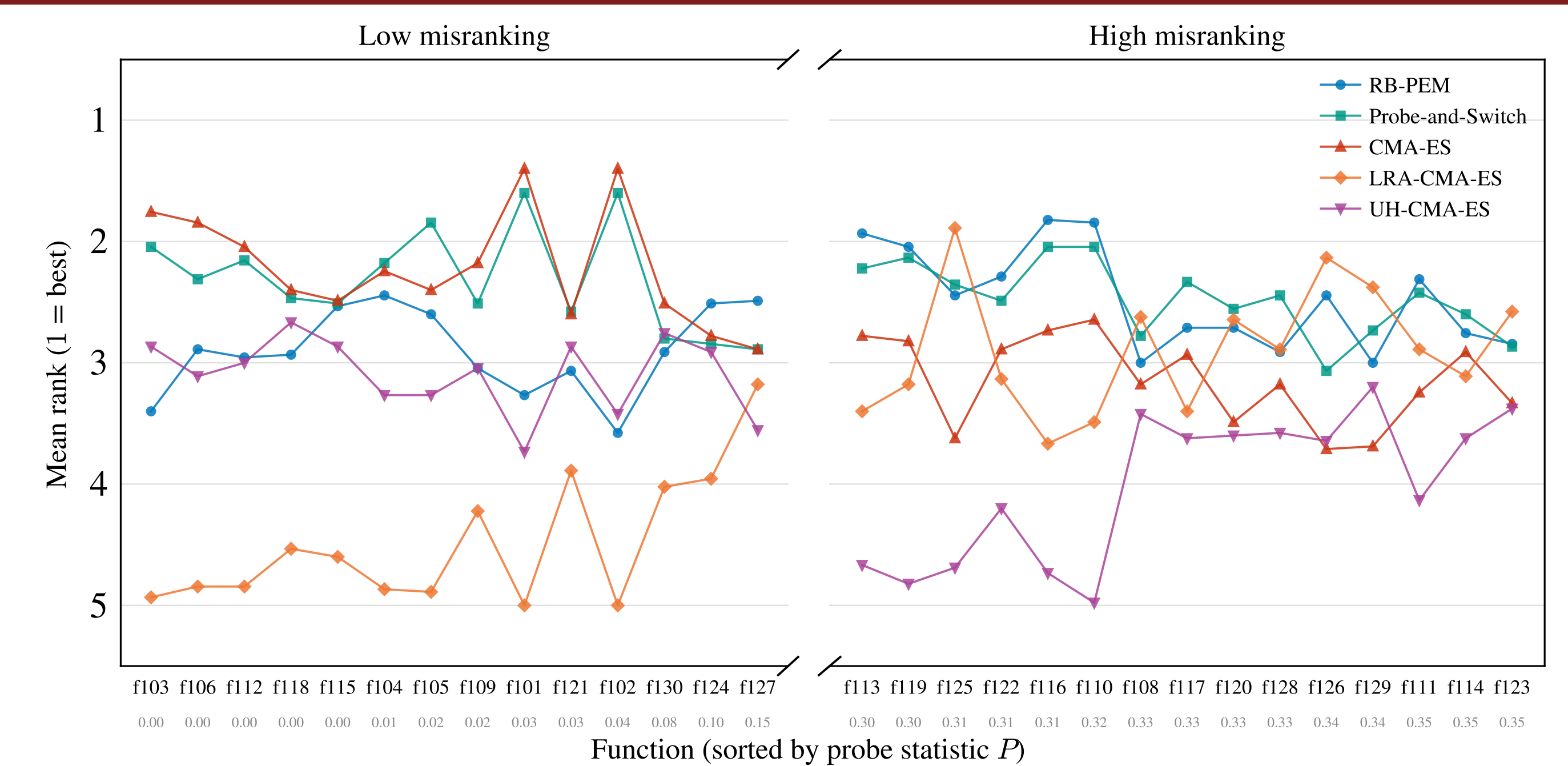
- Theorem 1: under local strong convexity, dispersion $\rightarrow$ unavoidable curvature loss
- “Ranking noise” $\rightarrow$ per-generation penalty; cut it per oracle call

Main Results — bbob-noisy



Fixed-budget convergence ($d=40, B=100d$): RB-PEM reaches lower regret while keeping near-maximal depth (right-side markers).

Bimodal Misranking



30 functions split into low / high-misranking clusters (gap 0.145, $\approx 3\times$) — not post-hoc.

Comprehensive & Takeaways

- Wins 77–86% vs evaluation-stage denoising
- Thresholds transfer to 9 targets
- Edge over CMA-ES grows with d : 230:210 $\rightarrow$ 257:170

High rank uncertainty? Integrate it into selection — don't pay evaluations to remove it.